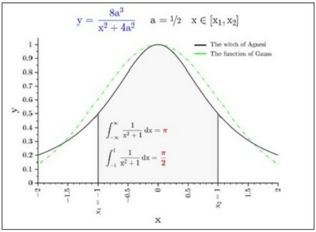

Figure 1: The witch of Agnesi and the function of Gauss

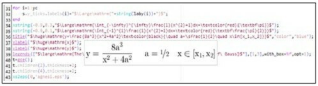

Figure 2: SciNotes preview

0	0	0	0	0	0	0	0
0.03	0.04	0.05	0.03	0.1	0.154	0.138	0.143
0.06	0.08	0.11	0.11	0.24	0.3	0.27	0.28
0.09	0.11	0.16	0.18	0.42	0.48	0.6	0.46
0.122	0.16	0.22	0.233	0.55	0.7	0.775	0.658
0.178	0.224	0.28	0.34	0.819	0.998	1.13	1.007
0.21	0.26	0.35	0.41	1	1.283	1.36	1.26
0.24	0.29	0.39	0.47	1.14	1.443	1.57	1.44
0.25	0.31	0.41	0.51	1.2	1.518	1.7	1.55
0.258	0.319	0.421	0.528	1.228	1.529	1.758	1.626
0.253	0.315	0.419	0.525	1.217	1.499	1.744	1.624
0.234	0.289	0.385	0.492	1.132	1.403	1.624	1.514
0.209	0.266	0.351	0.444	1.034	1.279	1.472	1.384
0.185	0.23	0.31	0.394	0.899	1.119	1.308	1.212
0.156	0.198	0.269	0.336	0.778	0.972	1.115	1.06
0.138	0.175	0.232	0.298	0.689	0.862	1.005	0.93
0.114	0.146	0.199	0.252	0.586	0.731	0.843	0.798
0.094	0.119	0.163	0.205	0.476	0.593	0.697	0.652
0.075	0.094	0.13	0.164	0.381	0.481	0.551	0.53
0.058	0.073	0.1	0.128	0.297	0.376	0.436	0.401
0.045	0.058	0.078	0.098	0.234	0.295	0.337	0.321
0.035	0.044	0.06	0.078	0.177	0.224	0.263	0.247
0	0	0	0	0	0	0	0

Figure 3: A data matrix in LaTeX

x_ticks.labels="$some-latex-here"+labx+"$".

5. **Y ticks labels.** In this example, they are numbers in a range between 0 and 1. The syntax is: *y_ticks.labels="$some-latex-here"+string(laby)+"$" If laby=string(0:0.1:1), the syntax is the same as in the X axis.*
6. **Legend.** This syntax is a bit long: *legends(["$first-row-text$";"$second-row-text$"],[first-row-color-number,second-row-color-number],with_box=%f,opt=1).* The option *with_box* can be *%f* (false) or *%t* (true) and *opt* is an integer in the range from 1 to 6.
7. **Two integrals under the curves.** Another example of an easy syntax: *xstring(x-position,y-position,"$some-latex-here$").* The area under the witch of Agnesi is equal to π, or π/2 under the grey part. In general, the area between the curve and the X axis is equal to 4 a^2 π (in the example shown, a=1/2).

The full code for this example *agnesi.sce* is available on GitHub.

From Scilab to LaTeX

In the May 2015 issue of *OSFY*, on page 103, there is a ribbon plot that shows one absorbance spectrum measured at eight different times. The data matrix is composed of eight columns and 23 rows. I can export this matrix directly to LaTeX using the function *prettyprint*. For the first four rows, the result is shown in the following code (the remaining rows are replaced by the dots):

```
${\begin{pmatrix}0&0&0&0&0&0&0&0\cr 0.03&0.04&0.05&0.03&0.1&0
.154&0.138&0.143\cr
0.06&0.08&0.11&0.11&0.24&0.3&0.27&0.28\cr 0.09&0.11&0.16&0.1
8&0.42&0.48&0.
6&0.46\cr ... \end{pmatrix}}$
```

Another way is to print the matrix in a graphic window with *xstring(0,0,prettyprint(data))*, as shown in Figure 3.

It's also possible to use Scilab as an almost automatic converter of tables from xls (via LibreOffice Calc) to LaTeX. The following code is composed of four parts: read the data from an xls table (text and numbers), get the maximum width of each column, print the column headers, and then print the data values. The caption and the tabular alignments are the only code rows that must be settled by hand. The original xls table is shown in Figure 4 (top); the output on Scilab Console is a table in LaTeX code that is fully formatted, as shown in Figure 4 (bottom). The data has been taken from Reference 4.

```
clear;
sheets=readxls(uigetfile("*.xls"));
data=sheets(1);
typeof(data);
data.text;
data.value;
data=string(data);
[nr,nc]=size(data);
// Max width of each column
w=zeros(nc,1);
for i=1:nc
    w(i)=max(length(data(:,i)));
end
printf(msprintf("\\\\begin{table}[htb]")+"\n"..
      +msprintf("\\\\caption{Physical properties of some
n-alkanes.}")+"\n"..
      +msprintf("\\\\centering")+"\n"..
      +msprintf("\\\\begin{tabular}{l|l|r|r|r}")+"\n");
// Columns headers
for i=1:nc
    printf("%"+string(w(i))+"s",data(1,i));
```